TIØ4145 Corporate Finance
Chapter 11

Felipe Van de Sande Araujo
Email: felipe.van.de.s.araujo@ntnu.no

Department of Industrial Economics and Technology Management
Norwegian University of Science and Technology
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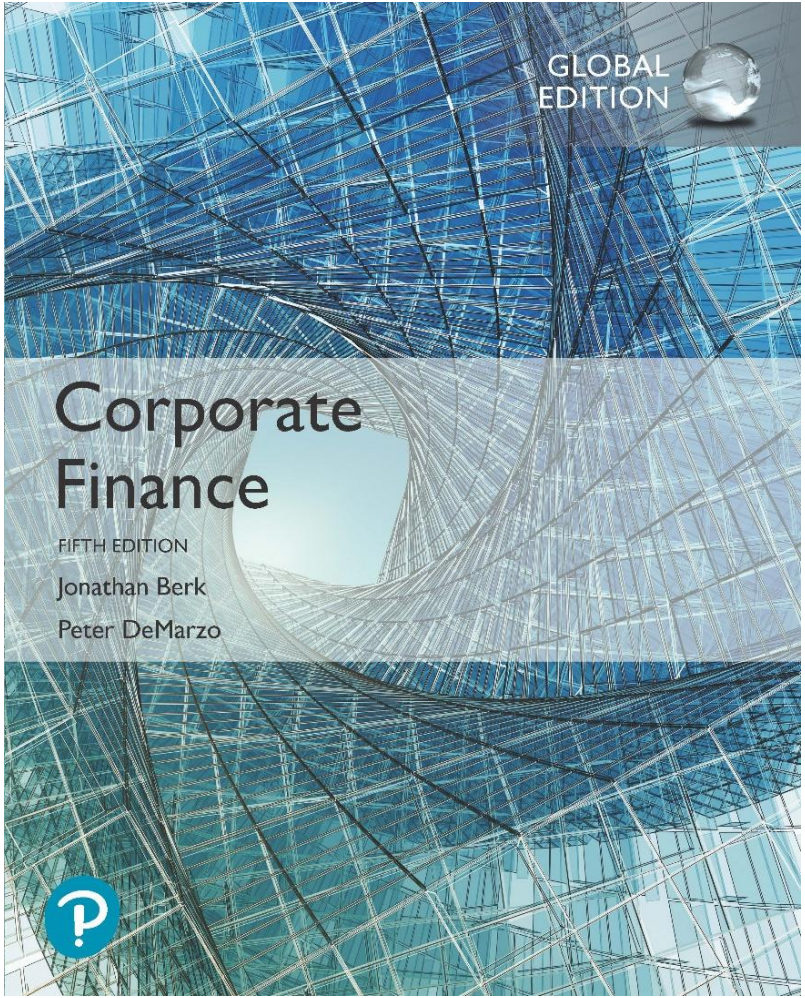

Chapter 11

Optimal Portfolio Choice and the Capital Asset Pricing Model

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Chapter Outline

11.1 The Expected Return of a Portfolio

11.2 The Volatility of a Two-Stock Portfolio

11.3 The Volatility of a Large Portfolio

11.4 Risk Versus Return: Choosing an Efficient Portfolio

11.5 Risk-Free Saving and Borrowing

11.6 The Efficient Portfolio and Required Returns

11.7 The Capital Asset Pricing Model

11.8 Determining the Risk Premium

Portfolio Weights

- The fraction of the total investment in the portfolio held in each individual investment in the portfolio
  - The portfolio weights must add up to 1.00 or 100%.

$$x_i = \frac{\text{Value of investment } i}{\text{Total value of portfolio}}$$

Return of a Portfolio

- The return of a portfolio, R_p, is the weighted average of the returns on the investments in the portfolio, where the weights correspond to portfolio weights.

$$R_p = x_1 R_1 + x_2 R_2 + \cdots + x_n R_n = \sum_{\mathrm{i}} x_i R_i$$

  - *n* investments.
  - x_n : portfolio weights.
  - R_n: return of the investment *n.*

The Expected Return of a Portfolio

- The expected return of a portfolio is the weighted average of the expected returns of the investments within it.

$$E[R_P] = E\left[\sum_i x_i R_i\right] = \sum_i E[x_i R_i] = \sum_i x_i E[R_i]$$

The Volatility of a Two-Stock Portfolio

Table 11.1 Returns and volatility for 3 stocks

	Stock Returns		
Year	North Air	West Air	Tex Oil
2013	21%	9%	−2%
2014	30%	21%	−5%
2015	7%	7%	9%
2016	−5%	−2%	21%
2017	−2%	−5%	30%
2018	9%	30%	7%
Average Return	10.0%	10.0%	10.0%
Volatility	13.4%	13.4%	13.4%

The Volatility of a Two-Stock Portfolio

Table 11.1 Returns and volatility for 3 stocks and portfolios of pairs

	Stock Returns			Portfolio Returns	
Year	North Air	West Air	Tex Oil	$1/2R_N + 1/2R_W$	$1/2R_W + 1/2R_T$
2013	21%	9%	−2%	15.0%	3.5%
2014	30%	21%	−5%	25.5%	8.0%
2015	7%	7%	9%	7.0%	8.0%
2016	−5%	−2%	21%	−3.5%	9.5%
2017	−2%	−5%	30%	−3.5%	12.5%
2018	9%	30%	7%	19.5%	18.5%
Average Return	10.0%	10.0%	10.0%	10.0%	10.0%
Volatility	13.4%	13.4%	13.4%	12.1%	5.1%

Diversification

- By combining stocks into a portfolio, we reduce risk through diversification.
- The amount of risk that is eliminated in a portfolio depends on the degree to which the stocks face common risks and their prices move together.

Risk of a Portfolio

- To find the risk of a portfolio we need to know more than the risk and return of the component stocks.

- How to express the degree to which the stocks' returns move together?

Covariance

- The expected product of the deviations of two returns from their means.
- Covariance between Returns R_i and R_j

$$Cov(R_i, R_j) = E[(R_i - E[R_i])\,(R_j - E[R_j])]$$

  - If the covariance is positive, the two returns tend to move together.
  - If the covariance is negative, the two returns tend to move in opposite directions.

- Estimate of the Covariance from Historical Data

$$Cov(R_i, R_j) = \frac{1}{T-1}\sum_t (R_{i,t} - \bar{R}_i)\,(R_{j,t} - \bar{R}_j)$$

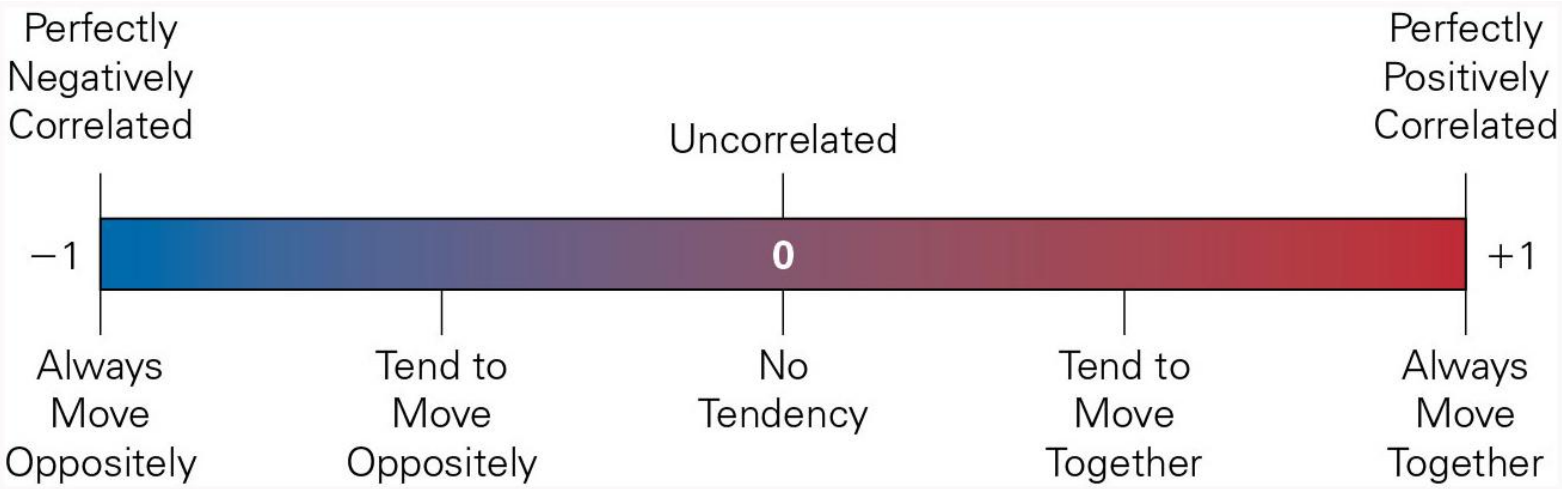
Correlation

- A measure of the common risk shared by stocks that does not depend on their volatility.

$$Corr(R_i, R_j) = \frac{Cov(R_i, R_j)}{SD(R_i)\, SD(R_j)}$$

  - The correlation has the same sign as the covariance.
  - The correlation between two stocks will always be between –1 and +1.

Correlation



Intuition

What are the covariance and the correlation of a stock's return with itself?

$$\mathrm{Cov}(R_i, R_j) = \mathrm{E}[(R_i - \mathrm{E}[R_i])\,(R_j - \mathrm{E}[R_j])]$$

$$\mathrm{Cov}(R_i, R_i) = \mathrm{E}[(R_i - \mathrm{E}[R_i])\,(R_i - \mathrm{E}[R_i])]$$

$$\mathrm{Cov}(R_i, R_i) = \mathrm{E}[(R_i - \mathrm{E}[R_i])^2] = \mathrm{Var}(R_i)$$

$$\mathrm{Corr}(R_i, R_j) = \frac{\mathrm{Cov}(R_i, R_j)}{\mathrm{SD}(R_i)\,\mathrm{SD}(R_j)}$$

$$\mathrm{Corr}(R_i, R_i) = \frac{\mathrm{Var}(R_i, R_i)}{SD(R_i)SD(R_i)} = 1$$

Computing the Covariance and Correlation Between Pairs of Stocks

Year	Deviation from Mean $(R_N - \overline{R}_N)$	$(R_W - \overline{R}_W)$	$(R_T - \overline{R}_T)$	North Air and West Air $(R_N - \overline{R}_N)(R_W - \overline{R}_W)$	West Air and Tex Oil $(R_W - \overline{R}_W)(R_T - \overline{R}_T)$
2013	11%	−1%	−12%	−0.0011	0.0012
2014	20%	11%	−15%	0.0220	−0.0165
2015	−3%	−3%	−1%	0.0009	0.0003
2016	−15%	−12%	11%	0.0180	−0.0132
2017	−12%	−15%	20%	0.0180	−0.0300
2018	−1%	20%	−3%	−0.0020	−0.0060
			Sum $= \sum_t (R_{i,t} - \overline{R}_i)(R_{j,t} - \overline{R}_j) =$	0.0558	−0.0642
Covariance:			$Cov(R_i, R_j) = \frac{1}{T-1}\text{Sum} =$	0.0112	−0.0128
Correlation:			$Corr(R_i, R_j) = \frac{Cov(R_i, R_j)}{SD(R_i)SD(R_j)} =$	0.624	−0.713

When will stock returns be more correlated?

	Microsoft	HP	Alaska Air	SW Airlines	Ford Motor	Kellogg	General Mills
Volatility:	32%	36%	36%	31%	47%	19%	17%
Correlation with:							
Microsoft	1						
HP	**0.40**	1					
Alaska Air	0.18	0.27	1				
Southwest Airlines	0.22	0.34	**0.40**	1			
Ford Motor	0.27	0.27	0.15	0.30	1		
Kellogg	0.04	0.10	0.15	0.15	0.17	1	
General Mills	0.10	0.06	0.20	0.21	0.08	**0.55**	1

Why is it that between different stocks we still find similar correlations?

Computing a Portfolio's Variance and Volatility

- How to calculate the variance of a two-stock portfolio?

$$\mathrm{Cov}(A+B, C) = \mathrm{Cov}(A, C) + \mathrm{Cov}(B, C)$$
and $\mathrm{Cov}(mA, B) = m\mathrm{Cov}(A, B)$

$$\begin{aligned}
\mathrm{Var}(R_P) &= \mathrm{Cov}(R_P, R_P) \\
&= \mathrm{Cov}(x_1R_1 + x_2R_2, x_1R_1 + x_2R_2) \\
&= x_1x_1\mathrm{Cov}(R_1, R_1) + x_1x_2\mathrm{Cov}(R_1, R_2) + x_2x_1\mathrm{Cov}(R_2, R_1) + x_2x_2\mathrm{Cov}(R_2, R_2) \\
&= x_1^2\mathrm{Var}(R_1) + 2x_1x_2\mathrm{Cov}(R_1, R_2) + x_2^2\mathrm{Var}(R_2)
\end{aligned}$$

- The variance and volatility of a two-stock portfolio:

$$\mathrm{Var}(R_P) = x_1^2\mathrm{Var}(R_1) + x_2^2\mathrm{Var}(R_2) + 2x_1x_2\mathrm{Cov}(R_1, R_2)$$

$$SD(R_P) = \sqrt{x_1^2\mathrm{SD}(R_1)^2 + x_2^2\mathrm{SD}(R_2)^2 + 2x_1x_2\mathrm{Cov}(R_1, R_2)}$$

The Volatility of a Large Portfolio

- The variance of a large portfolio can be calculated as:

$$\mathrm{Var}(R_P) = \sum_i x_i \mathrm{Cov}\,(R_i, R_P) = \sum_i x_i \mathrm{Cov}\left(R_i, \sum_j x_j R_j\right)$$
$$= \sum_i \sum_j x_i x_j \mathrm{Cov}\,(R_i, R_j)$$

- This can also be expressed as the weighted average covariance of each stock with the portfolio:

$$\mathrm{Var}(R_P) = \mathrm{Cov}\,(R_P, R_P) = \mathrm{Cov}\left(\sum_i x_i R_i, R_P\right) = \sum_i x_i \mathrm{Cov}(R_i, R_P)$$

$$\frac{\mathrm{Var}(R_P)}{\mathrm{SD}(R_P)} = \sum_i \frac{x_i \mathrm{Cov}(R_i, R_P)}{\mathrm{SD}(R_P)} \qquad \mathrm{SD}(R_P) = \sum_i x_i \mathrm{SD}(R_i) \mathrm{Corr}(R_i, R_P)$$

Diversification with General Portfolios

- For a portfolio with arbitrary weights, the standard deviation is calculated as:

$$SD(R_P) = \sum_i \overbrace{\underbrace{x_i}_{\substack{\text{Amount} \\ \text{of } i \text{ held}}} \times \underbrace{SD(R_i)}_{\substack{\text{Total} \\ \text{risk of } i}} \times \underbrace{Corr(R_i, R_p)}_{\substack{\text{Fraction of } i\text{'s} \\ \text{risk that is} \\ \text{common to } p}}}^{\substack{\text{Security } i\text{'s contribution to the} \\ \text{volatility of the portfolio}}}$$

- The risk of the portfolio will usually be lower than the weighted average volatility of the individual stocks:

$$SD(R_P) = \sum_i x_i SD(R_i)\, Corr(R_i, R_p) < \sum_i x_i SD(R_i)$$

Diversification with an Equally Weighted Portfolio

- Equally Weighted: A portfolio in which the same amount is invested in each stock.

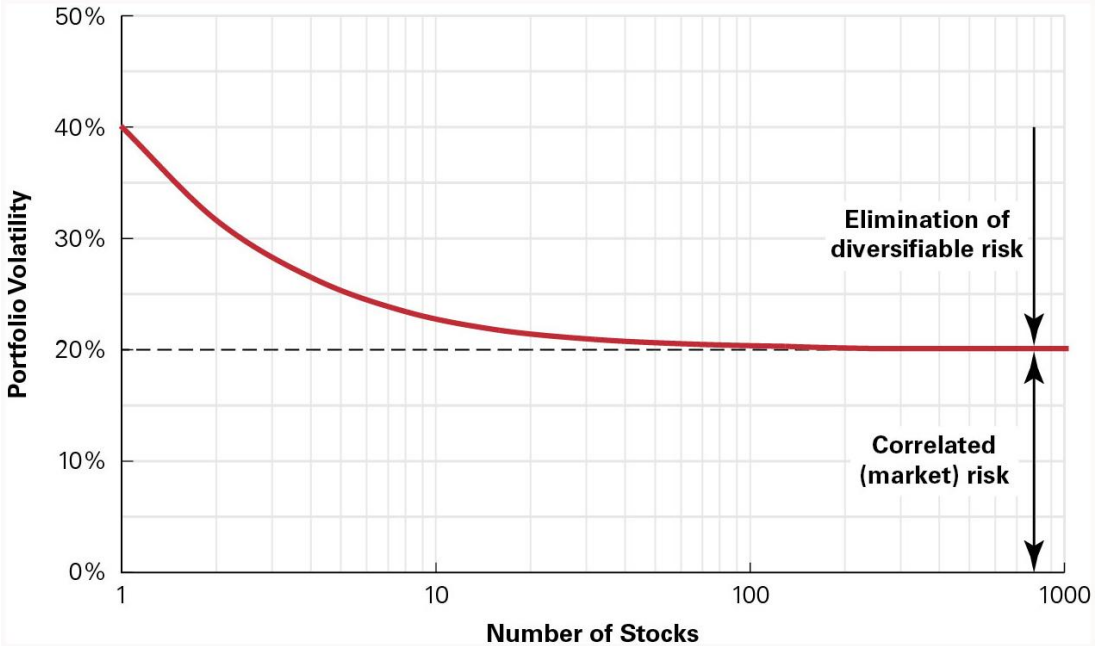
  $x = 1/n$

- Variance of an Equally Weighted Portfolio of *n* Stocks:

$$\mathrm{Var}(R_P) = \frac{1}{n}\underbrace{\mathrm{E}[\mathrm{Cov}(Ri, Rj)|i = j]}_{\text{Average Variance of the Individual Stocks}} + \left(1 - \frac{1}{n}\right)\underbrace{\mathrm{E}[\mathrm{Cov}(Ri, Rj)|i \neq j]}_{\text{Average Covariance between the Stocks}}$$

  - What happens when *n* is very large?

Volatility of an Equally Weighted Portfolio Versus the Number of Stocks



Efficient Portfolios with Two Stocks

- Consider a portfolio of Intel and Coca-Cola.
- Suppose the stocks have a correlation of 0.

	Expected Return	Volatility
Intel	26%	50%
Coca-Cola	6%	25%

- How should an investor choose a portfolio?
- Are some portfolios preferable?

Choosing an efficient portfolio

- Identifying Inefficient Portfolios
  - In an **inefficient portfolio**, it is possible to find another portfolio that is better in terms of both expected return and volatility.

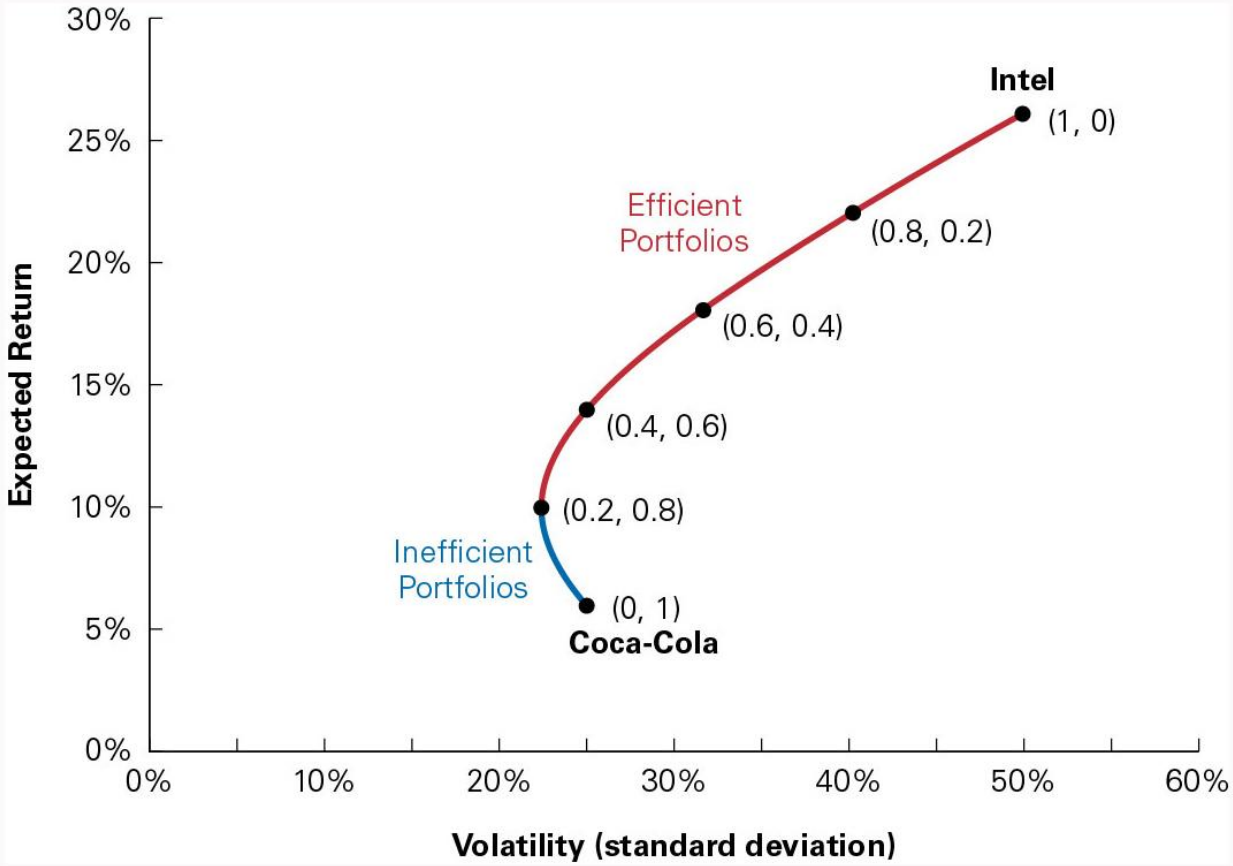
- Identifying Efficient Portfolios
  - In an **efficient portfolio** there is no way to reduce the volatility of the portfolio without lowering its expected return.

Expected Returns and Volatility for Different Portfolios of Two Stocks

Intel and Coca-Cola

Portfolio Weights		Expected Return (%)	Volatility (%)
x_I	x_C	$E[R_P]$	$SD[R_P]$
1.00	0.00	26.0	50.0
0.80	0.20	22.0	40.3
0.60	0.40	18.0	31.6
0.40	0.60	14.0	25.0
0.20	0.80	10.0	22.4
0.00	1.00	6.0	25.0

Volatility Versus Expected Return for Portfolios

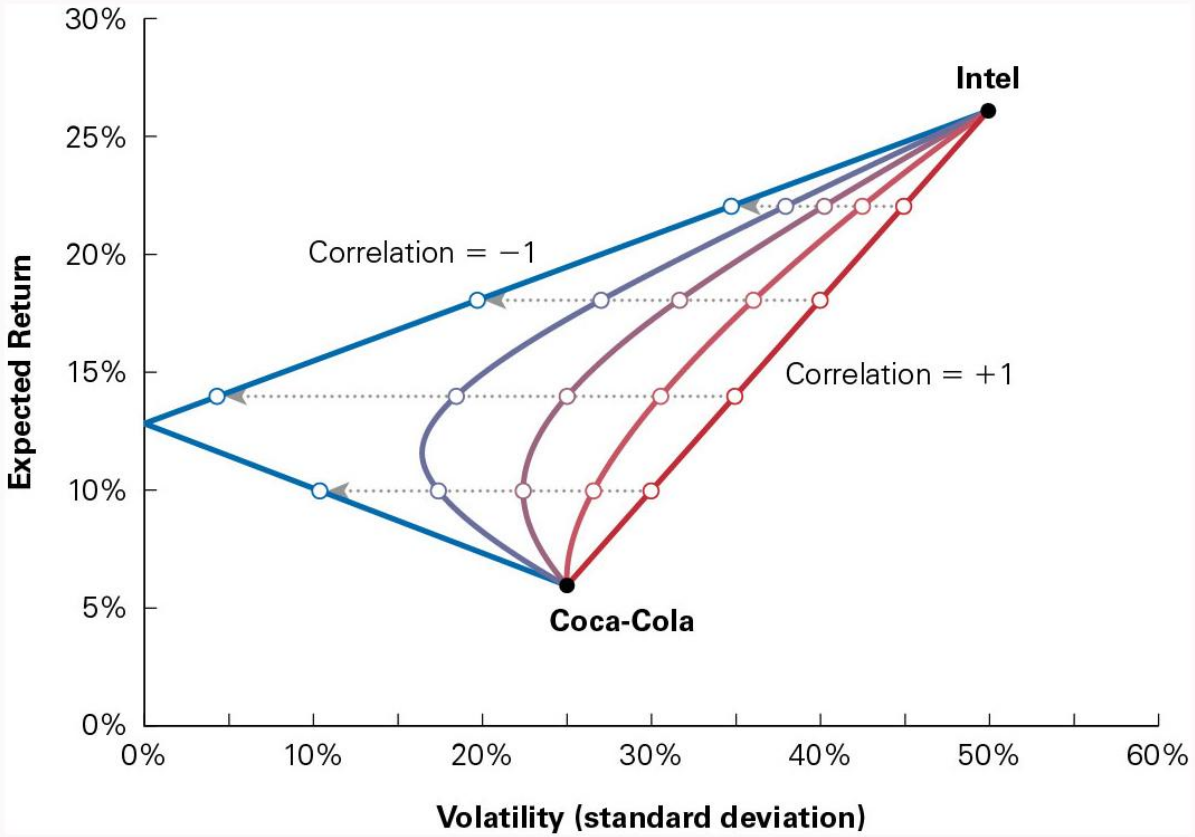


The Effect of Correlation

- Correlation has no effect on the expected return of a portfolio. However, the volatility of the portfolio will differ depending on the correlation.

- The lower the correlation, the lower the volatility we can obtain. As the correlation decreases, the volatility of the portfolio falls.

Effect on Volatility and Expected Return of Changing the Correlation

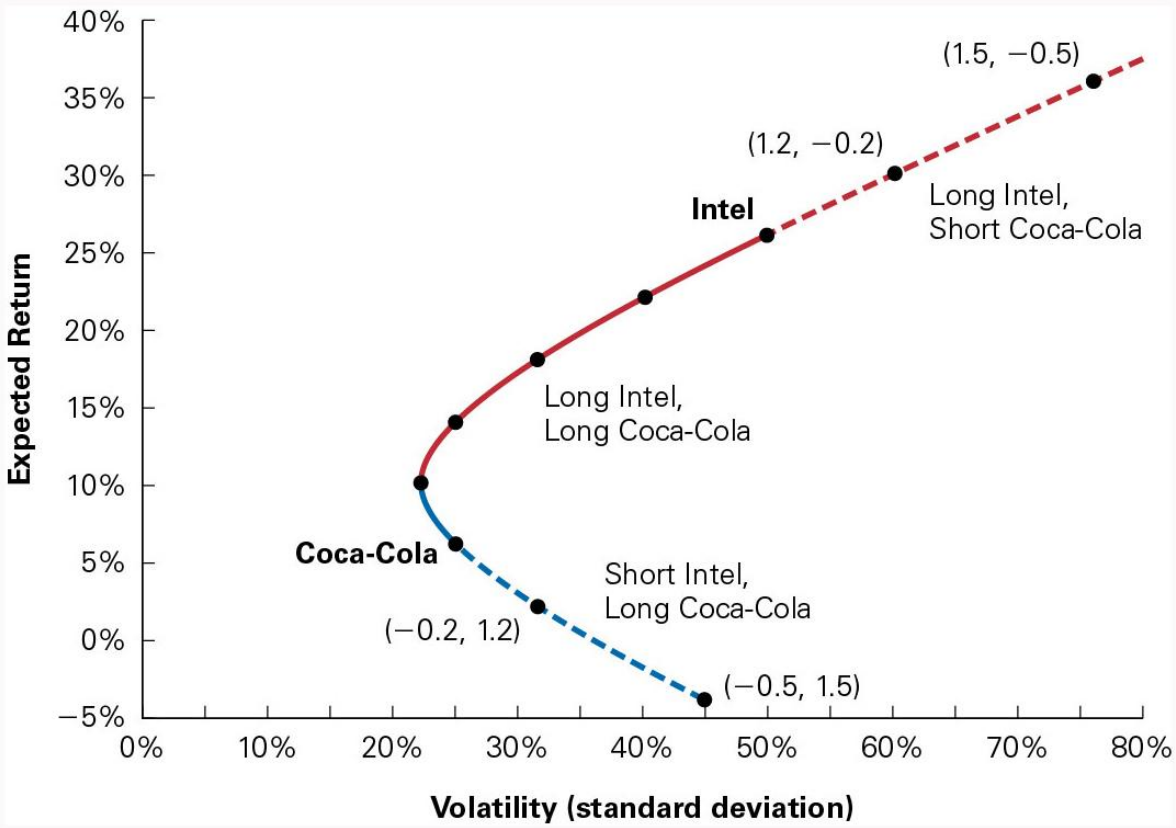


Are there other possible portfolios with only two assets?

Short Sales

- Long Position
  - A positive investment in a security
- Short Position
  - A negative investment in a security
  - In a short sale, you sell a stock that you do not own and then buy that stock back in the future.
  - Short selling is an advantageous strategy if you expect a stock price to decline in the future.

Portfolios of Intel and Coca-Cola Allowing for Short Sales



Example

Suppose you have $20,000 in cash to invest. You decide to short sell $10,000 worth of Coca-Cola stock and invest the proceeds from your short sale, plus your $20,000, in Intel. Assume that the stocks are uncorrelated.

How can we calculate the expected return and volatility of your portfolio?

$$x_1 = \frac{\text{Value of investment in Intel}}{\text{Total value of portfolio}} = \frac{30{,}000}{20{,}000} = 150\%$$

$$x_C = \frac{\text{Value of investment in Coca Cola}}{\text{Total value of portfolio}} = \frac{-10{,}000}{20{,}000} = -50\%$$

$$E[R_p] = x_1 E[R_1] + x_C E[R_C] = 1.50 \times 26\% + (-0.50) \times 6\% = 36\%$$

$$SD(R_P) = \sqrt{Var(R_P)} = \sqrt{x_1^2 Var(R_1) + x_C^2 Var(R_C) + 2x_1 x_C Cov(R_1, R_C)}$$

$$= \sqrt{1.5^2 \times 0.50^2 + (-0.5)^2 \times 0.25^2 + 2(1.5)(-0.5)(0)} = 76.0\%$$

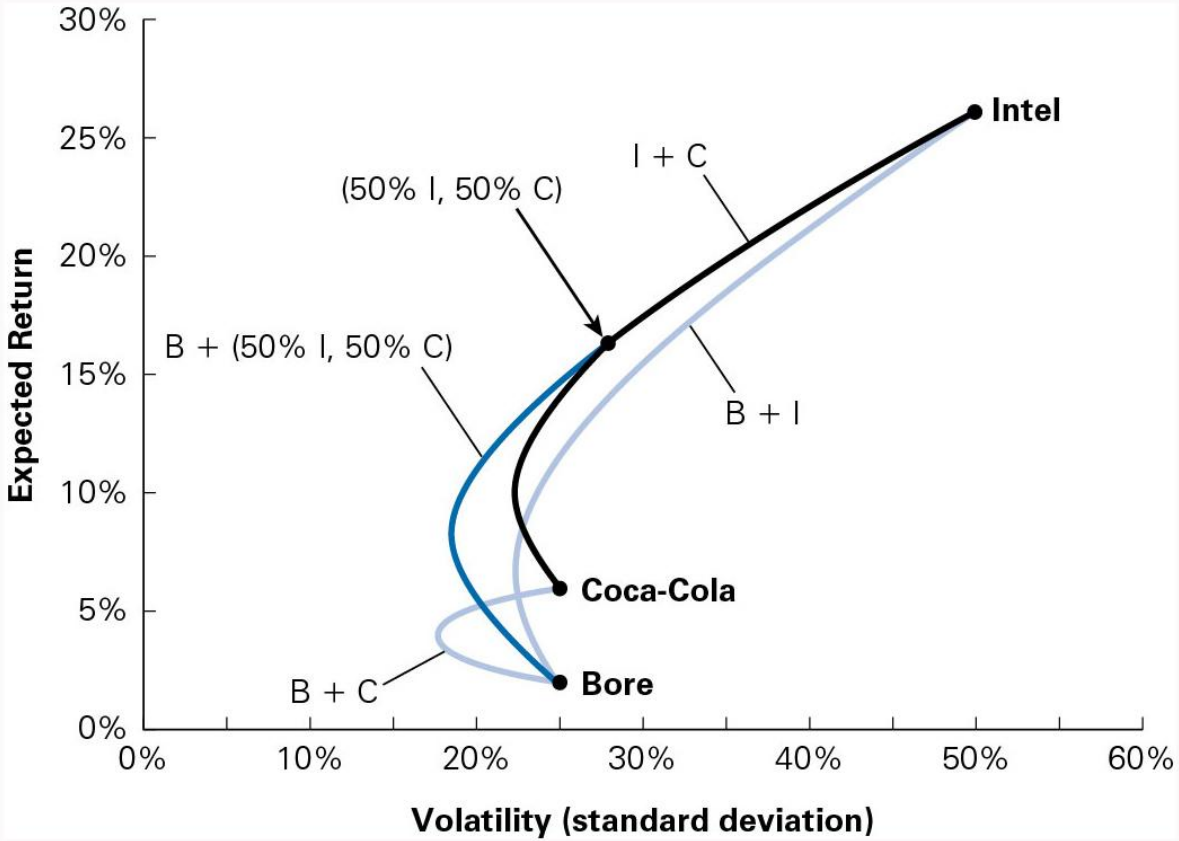
Efficient Portfolios with Many Stocks

- Consider adding Bore Industries to the two-stock portfolio:

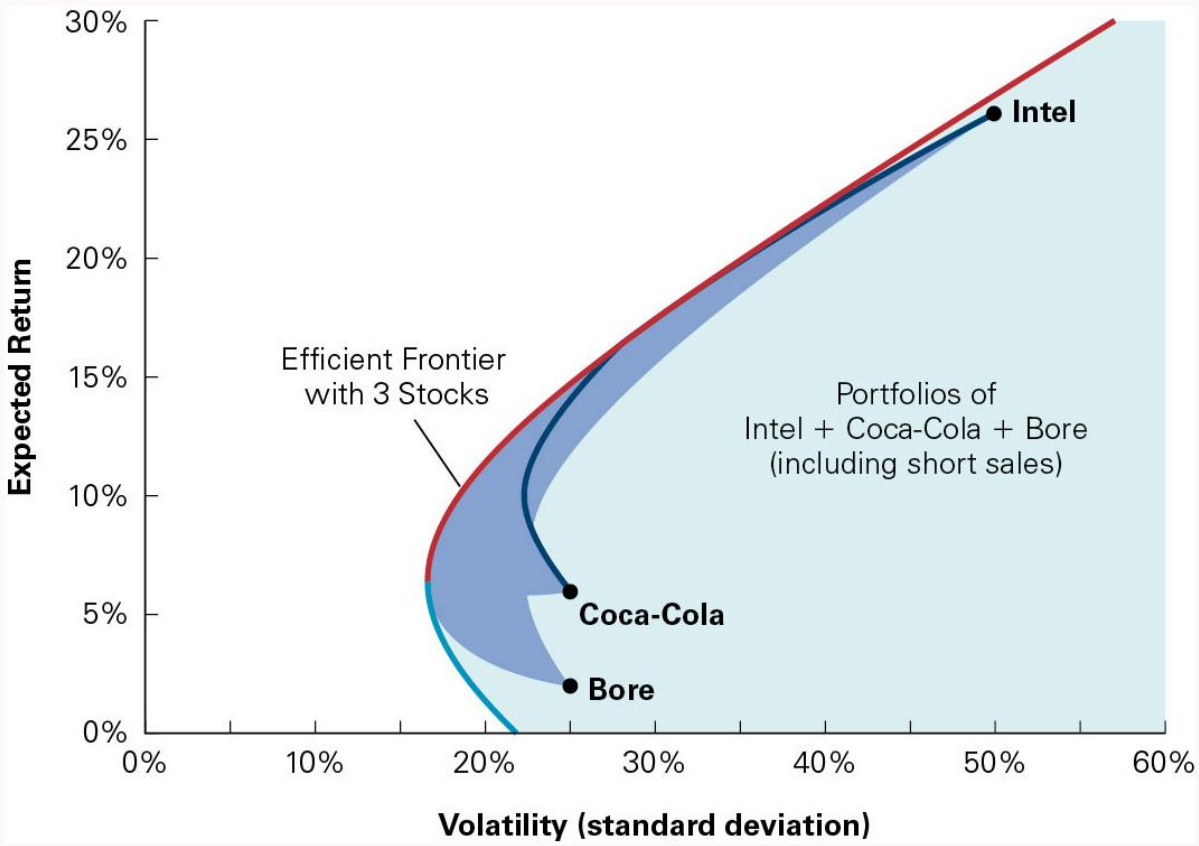
			Correlation with		
Stock	**Expected Return**	**Volatility**	**Intel**	**Coca-Cola**	**Bore Ind.**
Intel	26%	50%	1.0	0.0	0.0
Coca-Cola	6%	25%	0.0	1.0	0.0
Bore Industries	2%	25%	0.0	0.0	1.0

- Although Bore has a lower return and the same volatility as Coca-Cola, it still may be beneficial to add Bore to the portfolio for the diversification benefits.

Expected Return and Volatility for Selected Portfolios of Intel, Coca-Cola, and Bore Industries



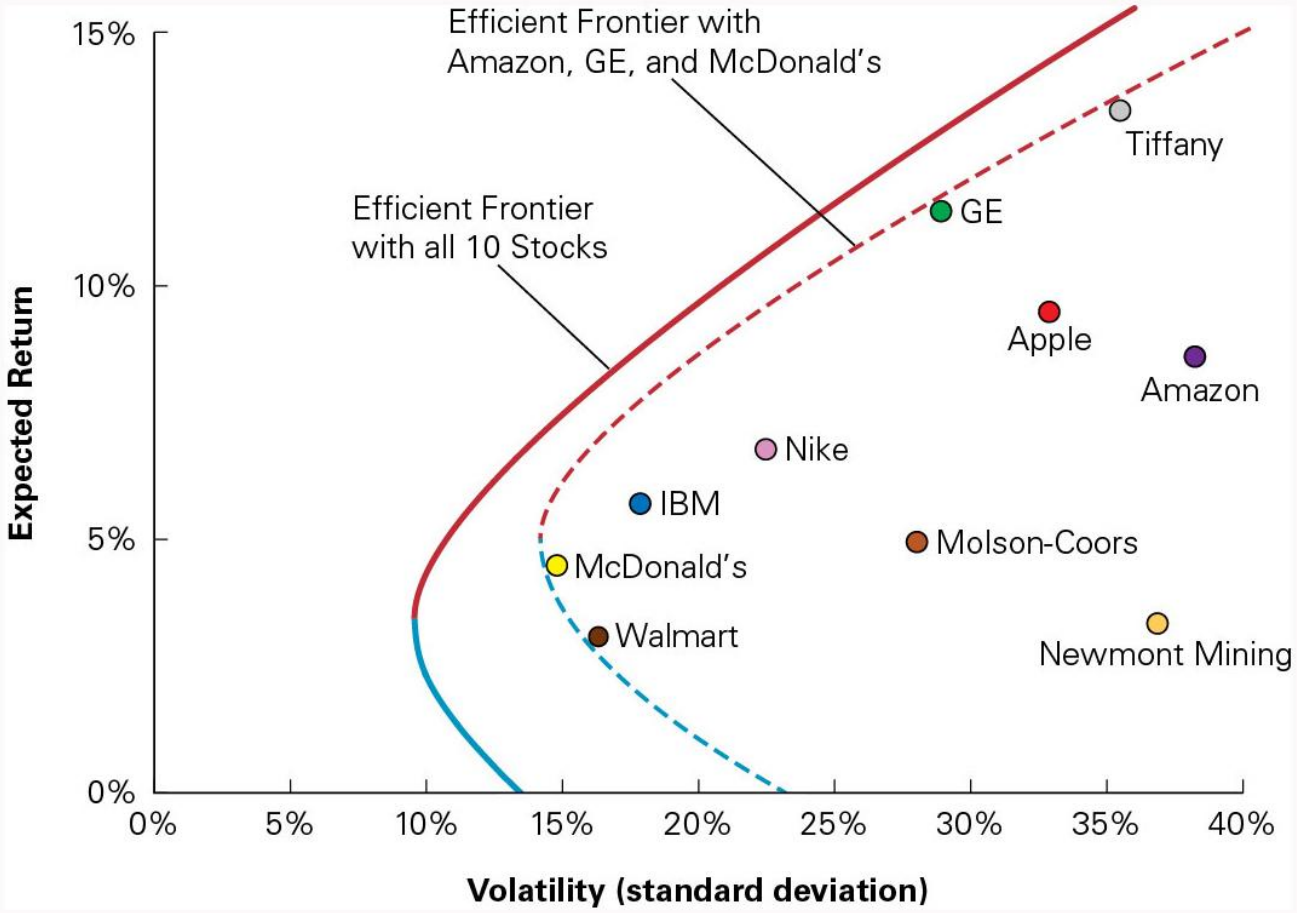
Volatility and Expected Return for All Portfolios of Intel, Coca-Cola, and Bore Industries



Risk Versus Return: Many Stocks

- The efficient portfolios are those on the northwest edge of the shaded region.
- This curve is called the **efficient frontier** for these three stocks.
- In the example, none of the stocks, on its own, is on the efficient frontier, so it would not be efficient to put all our money in a single stock.

Efficient Frontier with 3 vs. 10 Stocks



Risk-Free Saving and Borrowing

- Risk can also be reduced by investing a portion of a portfolio in a risk-free investment, like Treasure Bills. Doing so will also likely reduce the expected return.
- Buying Stocks on Margin: An aggressive investor who is seeking high expected returns might decide to borrow money to invest even more in the stock market.
- A portfolio that consists of a short position in the risk-free investment is known as a **levered portfolio**.
  - Margin investing is a risky investment strategy.

Investing in Risk-Free Securities

- Consider an arbitrary risky portfolio.
- What is the effect on risk and return of putting a fraction (x) of the money in the portfolio, while leaving the remaining fraction in risk-free Treasury bills?

$$\mathrm{E}\,[R_{xP}\,] = (1 - x)r_f + x\mathrm{E}[R_P] = r_f + x(\mathrm{E}[R_P] - r_f\,)$$

$$\begin{aligned} SD(R_{xP}) &= \sqrt{(1-x)^2 Var(r_f) + x^2 Var(R_P) + 2(1-x)x Cov(r_f, R_P)} \\ &= \sqrt{x^2 Var(R_P)} \\ &= x\,SD(R_P) \end{aligned}$$

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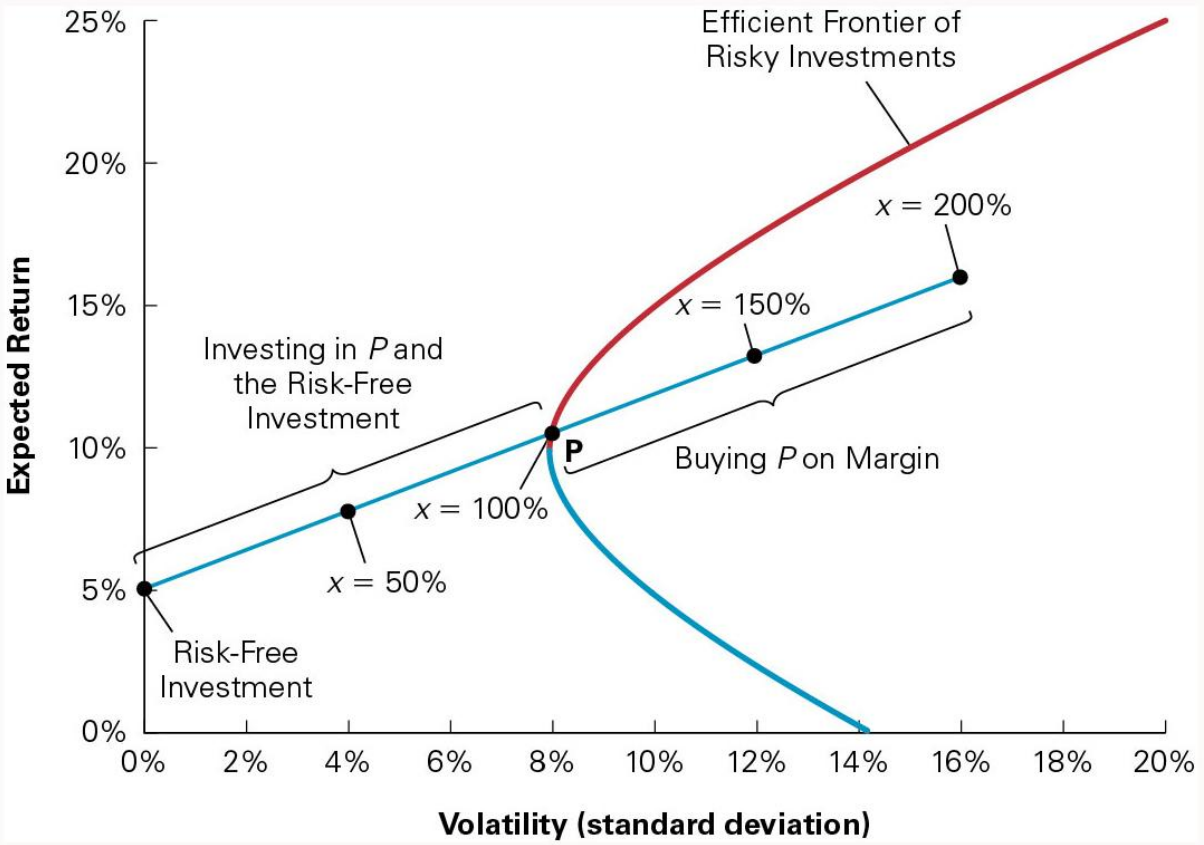
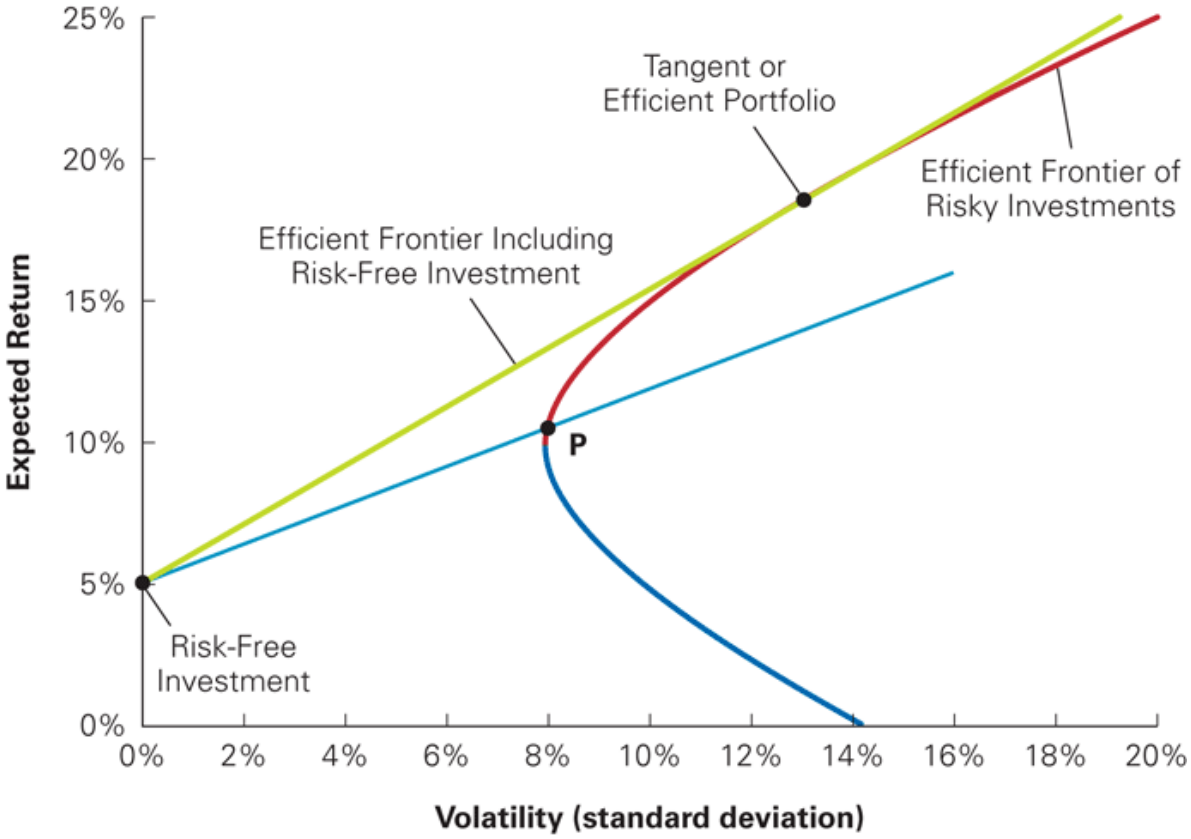
The Risk–Return Combinations from Combining a Risk-Free Investment and a Risky Portfolio



What does the slope of the line represent?

The Tangent or Efficient Portfolio



Identifying the Tangent Portfolio

- **Sharpe Ratio:**
  - Measures the ratio of reward-to-volatility provided by a portfolio.

$$\text{Sharpe Ratio} = \frac{\text{Portfolio Excess Return}}{\text{Portfolio Volatility}} = \frac{E[R_P] - r_f}{SD(R_P)}$$

- The portfolio with the highest Sharpe ratio is the portfolio where the line with the risk-free investment is tangent to the efficient frontier of risky investments.
- The portfolio that generates this tangent line is known as the **tangent portfolio**.

Risk and Return

- Combinations of the risk-free asset and the tangent portfolio provide the best risk and return trade-off available to an investor.
- This means that:
  - The tangent portfolio is efficient.
  - All efficient portfolios are combinations of the risk-free investment and the tangent portfolio.
  - Every investor should invest in the tangent portfolio **independent of his or her taste for risk**.

Investors preferences

- An investor's preferences will determine only **how much** to invest in the tangent portfolio versus the risk-free investment.
- Conservative investors will invest a small amount in the tangent portfolio.
- Aggressive investors will invest more in the tangent portfolio.
- Both types of investors will choose to hold the same portfolio of risky assets, the tangent portfolio, which is the **efficient portfolio**.

Incremental volatility from an investment

- Assume there is a portfolio of risky securities, *P*
- To determine whether *P* has the highest possible Sharpe ratio, consider whether its Sharpe ratio could be raised by adding more of some investment *i* to the portfolio.
- The contribution of investment *i* to the volatility of the portfolio depends on the risk that *i* has in common with the portfolio.

$$\text{Incremental volatility} = \text{SD}(R_i) \times \text{Corr}(R_i, R_P)$$

Improving a portfolio by adding an investment

- If you were to purchase more of investment *i* by borrowing, you would earn the expected return of *i* minus the risk-free return.
- Thus adding *i* to the portfolio *P* will improve our Sharpe ratio if:

$$\underbrace{E[R_i] - r_f}_{\text{Additional return from investment i}} > \underbrace{SD(R_i) \times Corr(R_i, R_P)}_{\text{Incremental volatility from investment i}} \times \underbrace{\frac{E[R_P] - r_f}{SD(R_P)}}_{\text{Return per unit of volatilty available from portfolio P}}$$

Portfolio Improvement: Beta and the Required Return

- Beta of investment *i* with Portfolio *P*

$$\beta_i^P = \frac{\mathrm{SD}(\mathrm{R_i}) \times \mathrm{Corr}(\mathrm{R_i}, \mathrm{R_P})}{\mathrm{SD}(\mathrm{R_P})} = \frac{\mathrm{Cov}(\mathrm{R_i}, \mathrm{R_P})}{\mathrm{Var}(\mathrm{R_P})}$$

- Increasing the amount invested in *i* will increase the Sharpe ratio of portfolio P if its expected return $\mathrm{E}[R_i]$ exceeds the **required return** r_i, which is given by:

$$r_i \equiv r_f + \beta_i^P \times (E[R_P] - r_f)$$

Example: Sharpe Ratio and Required Return for Different weights in Investment *i*

x_i	$E[R_p]$	$SD(R_p)$	Sharpe Ratio	Corr (R_i, R_p)	β_i^p	Required return r_i
0%	15.00%	20.00%	0.6000	10.0%	0.18	5.10%
4%	15.24%	20.19%	0.6063	16.8%	0.29	6.57%
8%	15.48%	20.47%	0.6097	23.4%	0.40	8.00%
10%	15.60%	20.65%	0.6103	26.6%	0.45	8.69%
11%	15.66%	20.74%	0.6104	28.2%	0.48	9.03%
12%	15.72%	20.84%	0.6103	29.7%	0.50	9.35%
16%	15.96%	21.30%	0.6084	35.7%	0.59	10.60%

Expected Returns and the Efficient Portfolio

- Relationship between the expected return of any security and its beta with the efficient portfolio:

$$E[R_i] = r_i \equiv r_f + \beta_i^{eff} \times (E[R_{eff}] - r_f)$$

- A portfolio is efficient if and only if the expected return of every available security equals its required return.
- We can determine the appropriate risk premium for an investment from its relative risk (beta) with the efficient portfolio.

The Capital Asset Pricing Model

- The Capital Asset Pricing Model (CAPM) allows us to identify the efficient portfolio of risky assets without having any knowledge of the expected return of each security.
- Instead, the CAPM uses the optimal choices investors make to identify the efficient portfolio as the market portfolio, the portfolio of all stocks and securities in the market.

The CAPM Assumptions

- Assumption 1: Investors can buy and sell all securities at competitive market prices (without incurring taxes or transaction costs) and can borrow and lend at the risk-free interest rate.
- Assumption 2: Investors hold only efficient portfolios of traded securities — portfolios that yield the maximum expected return for a given level of volatility.
- Assumption 3: Investors have homogeneous expectations regarding the volatilities, correlations, and expected returns of securities.
  - Homogeneous Expectations: all investors have the same estimates concerning future investments and returns.

Supply, Demand, and the Efficiency of the Market Portfolio

- Given homogeneous expectations, all investors will demand the same efficient portfolio of risky securities.
- The combined portfolio of risky securities of all investors must equal the efficient portfolio.
- Thus, if all investors demand the efficient portfolio, and the supply of securities is the market portfolio, the demand for market portfolio must equal the supply of the market portfolio.

Exercise

Suppose that after much research, you have identified the efficient portfolio. As part of your holdings, you have decided to invest $10,000 in Microsoft, and $5000 in Pfizer stock. Your friend, who is a more conservative investor, has $2000 invested in Pfizer. If your friend's portfolio is also efficient, how much has she invested in Microsoft?

a. $1,000 b. $4,000 c. Depend on risk profile.

Solution

b. $4,000

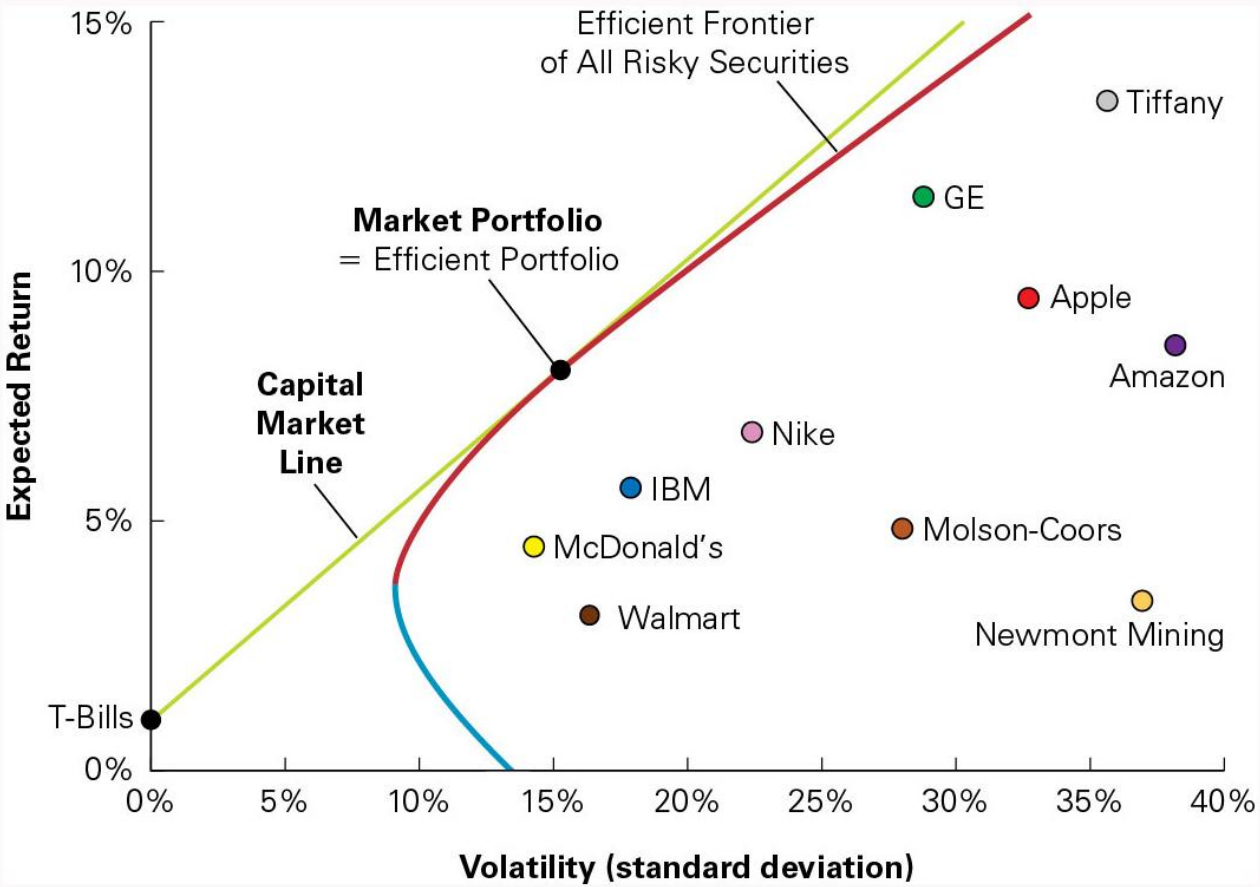
Because all efficient portfolios are combination of the risk-free investment and the tangent portfolio, they share the same proportions of risky stocks. Thus, since you have invested twice as much in Microsoft as in Pfizer, the same must be true for your friend; therefore, she has invested $4000 in Microsoft stock.

If all investors hold efficient portfolios, the same must be true of each of their portfolios. Because, collectively, all investors own all shares of Microsoft and Pfizer, Microsoft's market capitalization must therefore be twice that of Pfizer's.

Optimal Investing: The Capital Market Line

- When the CAPM assumptions hold, an optimal portfolio is a combination of the risk-free investment and the market portfolio.
- When the tangent line goes through the market portfolio, it is called the **capital market line (CML)**.

The Capital Market Line



Optimal Investing: The Capital Market Line

- The expected return and volatility of a capital market line portfolio are:

$$E[R_{xCML}] = (1-x)r_f + xE[R_{Mkt}] = r_f + x(E[R_{Mkt}] - r_f)$$

$$SD(R_{xCML}) = xSD(R_{Mkt})$$

Determining the Risk Premium

- Market Risk and Beta
  - Given an efficient market portfolio, the expected return of an investment is:

$$E[R_i] = r_i = r_f + \underbrace{\beta_i \left(E[R_{Mkt}] - r_f\right)}_{\text{Risk premium for security } i}$$

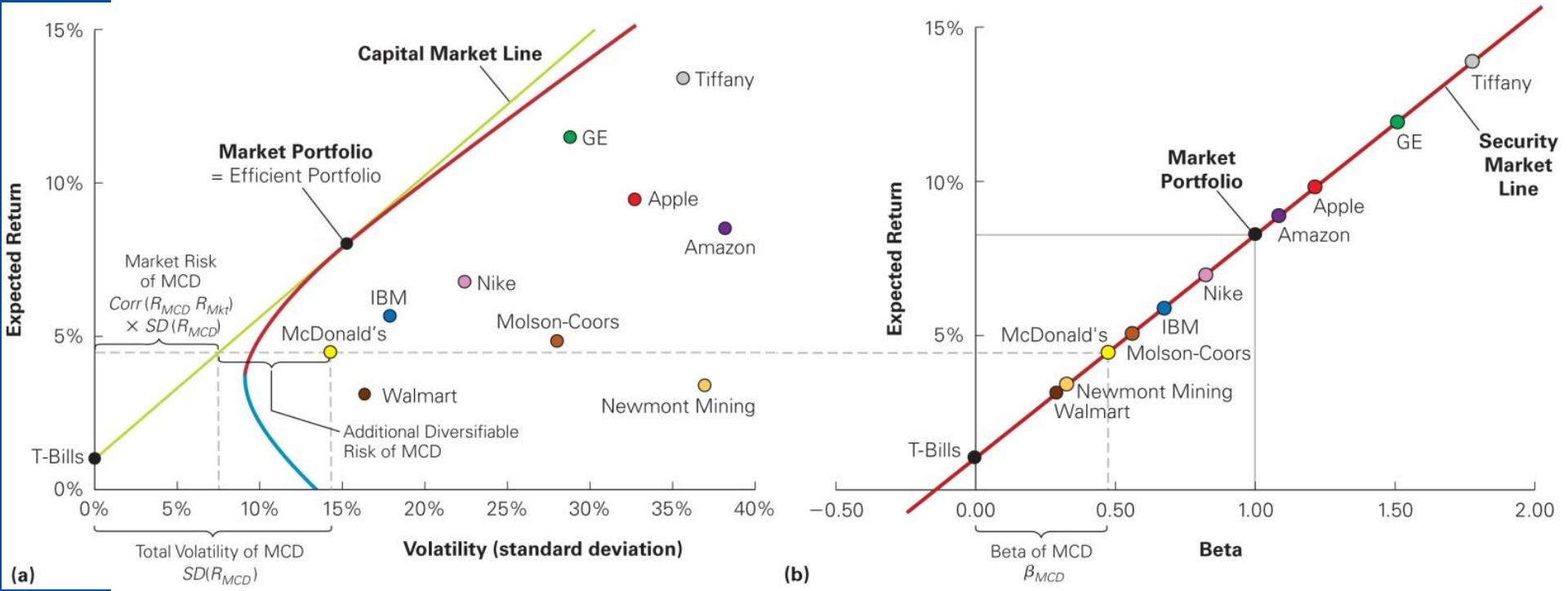
  - The beta is defined as follows:

$$\beta_i = \frac{\overbrace{SD(R_i) \times Corr(R_i, R_{Mkt})}^{\text{Volatility of } i \text{ that is common with the market}}}{SD(R_{Mkt})} = \frac{Cov(R_i, R_{Mkt})}{Var(R_{Mkt})}$$

The Security Market Line

- There is a linear relationship between a stock's beta and its expected return (See figure on next slide).
- The **security market line (SML)** is graphed as the line through the risk-free investment and the market.
  - According to the CAPM, if the expected return and beta for individual securities are plotted, they should all fall along the SML.

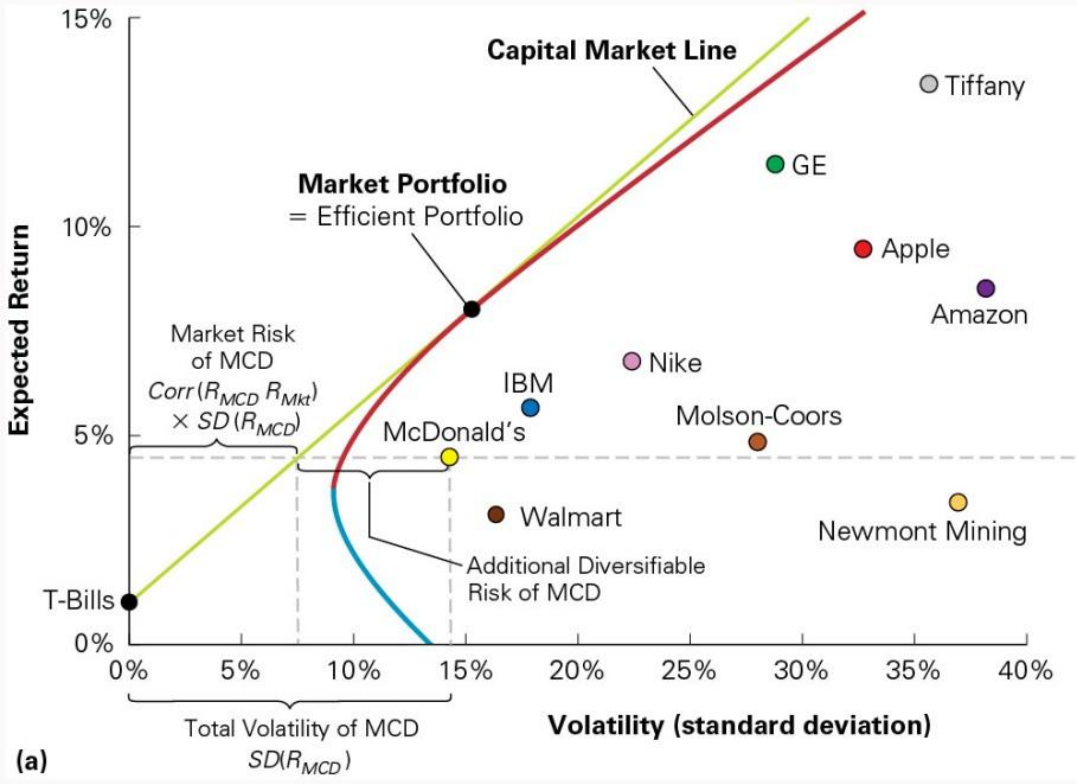
The Capital Market Line and the Security Market Line



The Capital Market Line, Panel (a)

The CML depicts portfolios combining the risk-free investment and the efficient portfolio, and shows the highest expected return that we can attain for each level of volatility.

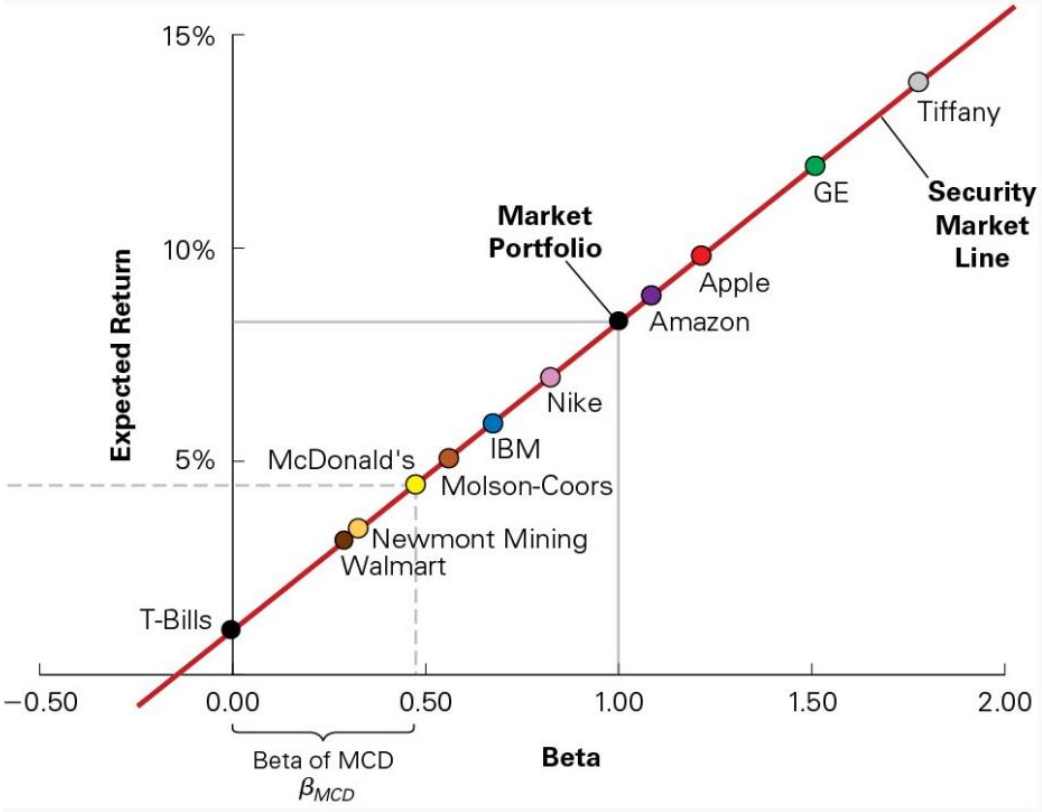
According to the CAPM, the market portfolio is on the CML and all other stocks and portfolios contain diversifiable risk and lie to the right of the CML, as illustrated for McDonald's.



The Security Market Line, Panel (b)

The SML shows the expected return for each security as a function of its beta with the market.

According to the CAPM, the market portfolio is efficient, so all stocks and portfolios should lie on the SML



Evaluating Portfolios within CAPM

- The beta of a portfolio is the weighted average beta of the securities in the portfolio.

$$\beta_P = \frac{Cov(R_P, R_{Mkt})}{Var(R_{Mkt})} = \frac{Cov\left(\sum_i x_i R_i, R_{Mkt}\right)}{Var(R_{Mkt})} = \sum_i x_i \frac{Cov(R_i, R_{Mkt})}{Var(R_{Mkt})} = \sum_i x_i \beta_i$$

Example

- Suppose the stock of the 3M Company (MMM) has a beta of 0.69 and the beta of Hewlett-Packard Co. (HPQ) stock is 1.77. Assume the risk-free interest rate is 5% and the expected return of the market portfolio is 12%.
- How can we calculate the expected return of a portfolio of 40% of 3M stock and 60% Hewlett-Packard stock, according to the CAPM?

$$\beta_P = \sum_i x_i \beta_i = (.40)(0.69) + (.60)(1.77) = 1.338$$

$$\mathrm{E}[R_P] = r_f + \beta_P\,(\mathrm{E}[R_{\mathrm{Mkt}}] - r_f)$$

$$\mathrm{E}[R_P] = 5\% + 1.338 \times (12\% - 5\%) = 14.37\%$$

Very short summary of the Capital Asset Pricing Model

- The market portfolio is the efficient portfolio.
- The risk premium for any security is proportional to its beta with the market.

Learning Objectives

- Describe the contribution of each security to the portfolio.
- Use the definition of an efficient portfolio to describe the efficient frontier.
- Explain the effect of combining a risk-free asset with a portfolio of risky assets.
- Describe what is meant by a short sale, and illustrate how short selling extends the set of possible portfolios.
- Explain how an individual investor will choose from the set of efficient portfolios.

Learning Objectives

- Beta for an individual stock and for a portfolio :
  - Use the risk-free rate, the expected return on a portfolio, and the beta of a security to decide whether buying shares of that security will improve the performance of the portfolio.
  - Use the risk-free rate, the expected return on the efficient portfolio, and the beta of a security with the efficient portfolio to calculate the risk premium for an investment.

- Sharpe ratio:
  - Define and explain how it helps identify the portfolio with the highest possible expected return for any level of volatility, and how this information can be used to identify the efficient portfolio.
  - Explain why the expected return must equal the required return.

Learning Objectives

- CAPM:
  - List the three main assumptions underlying the Capital Asset Pricing Model.
  - Explain why the CAPM implies that the market portfolio of all risky securities is the efficient portfolio.
  - Compare and contrast the capital market line with the security market line.

Learning Objectives

- Calculate:
  - Portfolio weight of each stock (Eq. 11.1)
  - Expected return on the portfolio (Eq. 11.3)
  - Covariance of each pair of stocks in the portfolio (Eq. 11.5)
  - Correlation coefficient of each pair of stocks in the portfolio (Eq. 11.6)
  - Variance of the portfolio (Eq. 11.8)
  - Standard deviation of the portfolio
  - Variance of an equally weighted portfolio (Eq. 11.12)

Thank you! Find me here:

Felipe Van de Sande Araujo

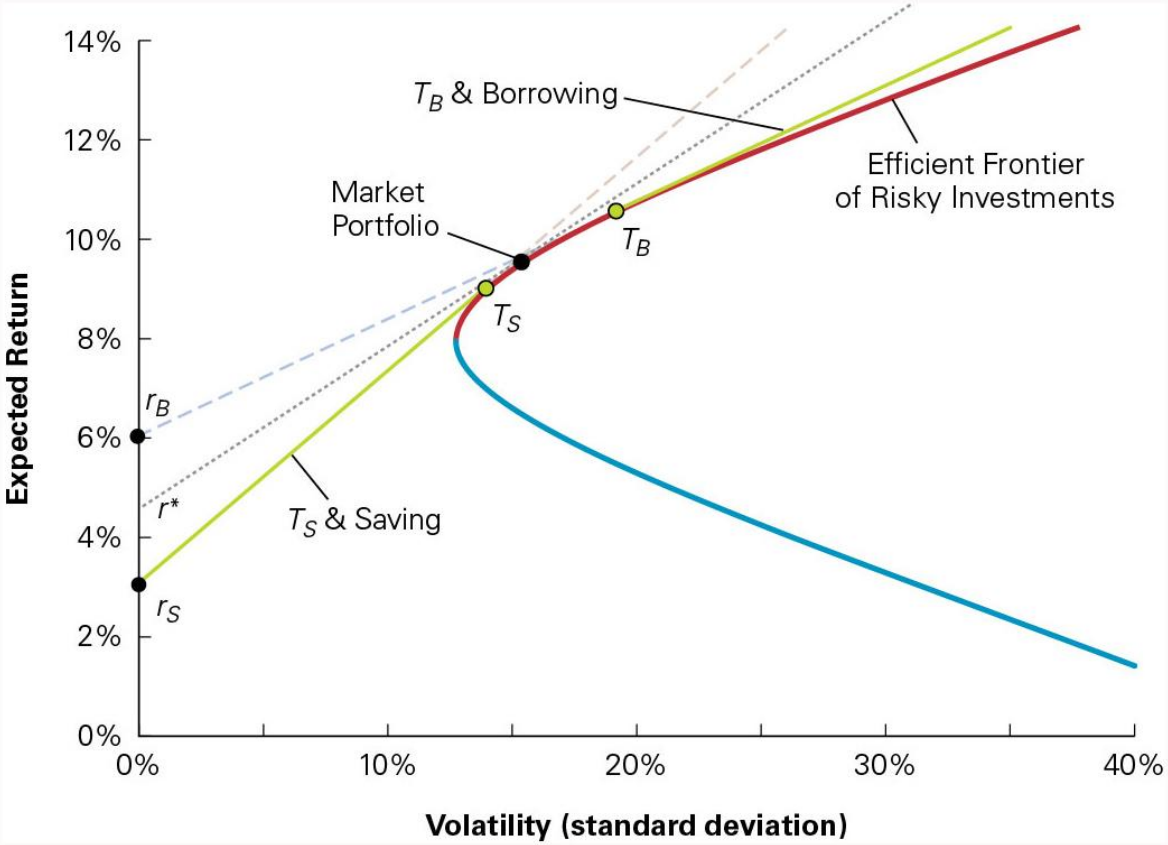
Sentralbygg 1, office 1147

felipe.van.de.s.araujo@ntnu.no

Appendix: The CAPM with Differing Interest Rates

- If borrowing and lending rates differ, then investors with different preferences will choose different portfolios of risky securities.
- With different saving and borrowing rates, the CAPM conclusion that the market portfolio is the unique efficient portfolio of risky investments is no longer valid.

The CAPM with Different Saving and Borrowing Rates



Appendix: The Security Market Line with Differing Interest Rates

- Determination of the security market line depends only on the market portfolio being tangent for some interest rate; the SML still holds in the following form:

$$E[R_i] = r^* + \beta_i \times (E[R_{Mkt}] - r^*)$$

- The SML holds with some rate r^* between r_S and r_B in place of r_f.